

Convolutional Neural Networks and Computer Vision

Luis Usero ¹, Juan P. Llerena ^{1,2} and Miguel A. Patricio ^{2,*}

¹ Cognitive Science Research Group, Universidad de Alcalá, 28805 Alcalá de Henares, Madrid, Spain; luis.usero@uah.es (L.U.); jp.llerena@uah.es (J.P.L.)

² Applied Artificial Intelligence Group, Computer Science and Engineering Department, Universidad Carlos III de Madrid (ROR Code 03ths8210), 28270 Colmenarejo, Madrid, Spain

* Correspondence: mpatrici@inf.uc3m.es

1. Introduction

The rapid development of artificial intelligence has transformed computer vision from a predominantly laboratory-oriented discipline into a broad family of deployable technologies for sensing, interpretation, decision support, and process automation. This evolution has been supported by substantial advances in convolutional neural networks (CNNs), transformer-based architectures, attention mechanisms, self-supervised representation learning, and task-specific optimization strategies. Rather than being displaced by newer architectural paradigms, convolutional models have continued to evolve through the incorporation of modern training procedures, large receptive fields, improved normalization mechanisms, and self-supervised pretraining. Representative developments such as ConvNeXt V2 have demonstrated that carefully redesigned convolutional architectures remain competitive across image classification, object detection, and semantic segmentation tasks [1]. Similarly, large-kernel convolutional networks have shown that CNN-based models can provide strong and transferable representations across multiple perception domains [2].

At the same time, the emergence of visual foundation models has considerably expanded the generalization capabilities of computer vision systems. Promptable models such as the Segment Anything Model have introduced a flexible formulation of image segmentation in which a single pretrained model can be transferred to previously unseen objects, image distributions, and application scenarios with limited or no task-specific retraining [3]. These developments are complemented by self-supervised and large-scale pretraining strategies, which reduce the dependence on exhaustively annotated datasets and facilitate the adaptation of visual representations to domains in which labeled data are scarce, expensive, or difficult to obtain.

Efficiency has also become a central requirement in contemporary computer vision. In many applied contexts, improvements in predictive accuracy are insufficient unless the resulting models can operate under strict constraints related to latency, memory consumption, energy usage, and hardware availability. Memory-efficient transformer architectures have therefore been developed to reduce computational redundancy while maintaining competitive recognition performance [4]. More recent models, such as SHViT, further simplify attention mechanisms and architectural design to improve the speed–accuracy trade-off on graphics processing units, central processing units, and mobile devices [5]. In parallel, state-space architectures such as Vision Mamba have emerged as an alternative to conventional self-attention, offering efficient long-range visual modeling and reduced memory requirements for high-resolution image processing [6].



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These advances are particularly relevant to *Applied Sciences*, where visual perception methods are expected not only to improve benchmark-level performance but also to address practical requirements associated with heterogeneous data, limited annotations, domain shifts, computational cost, interpretability, reliability, and integration into operational workflows. In this context, the practical value of a computer vision system depends on its capacity to maintain stable performance under real acquisition conditions, to provide predictions within the time constraints of the target application, and to support decisions made by domain specialists. Consequently, contemporary research increasingly combines architectural innovation with domain-specific knowledge, lightweight inference, uncertainty analysis, explainability mechanisms, and rigorous validation under representative operating conditions.

This Special Issue brings together ten contributions that illustrate the breadth of contemporary applied computer vision. The included studies address unmanned aerial vehicle detection in thermal imagery, road extraction from satellite images, gluten detection in bread images, social-emotional music classification, glioblastoma detection from magnetic resonance imaging, wall moisture detection in heritage sites using infrared imagery, lightweight self-supervised monocular depth estimation, steel surface defect detection, rear vehicle signal recognition for advanced driver assistance systems, and robust albedo-guided monocular depth estimation. Collectively, these contributions demonstrate how learning-based methods can be adapted to highly diverse data modalities, including RGB images, thermal imagery, satellite imagery, magnetic resonance images, musical features, industrial surface images, and video-derived monocular visual streams.

Despite their diversity, the studies share several methodological concerns that characterize current research in applied computer vision. These include the preservation of fine-grained spatial information, the extraction of multi-scale representations, the reduction of model complexity, robustness against acquisition variability, the mitigation of dataset bias, and the development of models capable of operating in real or near-real time. The contributions also reflect a broader transition from generic benchmark-oriented architectures toward solutions specifically designed around the physical, computational, and organizational constraints of each application domain.

The purpose of this editorial is to provide an overview of the Special Issue, identify the thematic and methodological connections among the published studies, and outline future directions for research in applied artificial intelligence and computer vision. Particular attention is given to the continuing relevance of convolutional architectures, their integration with attention and self-supervised learning mechanisms, and the increasing importance of efficiency, robustness, generalization, and trustworthy deployment.

2. An Overview of Thematic Contributions

The contributions to this Special Issue can be organized into four main thematic clusters. The first cluster concerns visual detection and recognition in safety-critical environments. This includes thermal drone detection for security applications, real-time rear vehicle signal detection for advanced driver assistance systems, and surface defect detection in industrial production. These papers emphasize fast inference, model robustness, and reliable recognition under challenging visual conditions.

The second cluster focuses on semantic extraction and scene understanding. Road extraction from satellite imagery and monocular depth estimation are representative of this direction. These studies show how model architectures can be redesigned to preserve spatial structure, extract multi-scale information, and obtain geometrically meaningful predictions while reducing computational requirements.

The third cluster addresses domain-specific decision support in health, food safety, cultural heritage, and inclusive education. Contributions in this cluster apply deep learning to gluten detection in bread images, glioblastoma detection from MRI, wall moisture detection in heritage sites, and social–emotional music classification for individuals with special needs. These studies illustrate the increasing importance of tailoring artificial intelligence systems to specific societal and professional needs, where performance must be considered alongside usability, fairness, accessibility, and contextual validity.

The fourth cluster relates to robustness, generalization, and efficient learning. Several contributions explicitly address common limitations of deep models, including limited data availability, domain shift, non-Lambertian illumination, edge-device constraints, and the need for lightweight inference. This concern is particularly evident in the contributions on self-supervised monocular depth estimation, meta-learning-enhanced object detection, demographic-bias analysis in medical imaging, and lightweight industrial inspection.

3. Summary of Contributions

Contribution 1 presents a comparative study of convolutional neural networks and transformer architectures for drone detection in thermal images. This contribution highlights the importance of matching model architecture to operational constraints in infrared security applications.

Contribution 2 proposes an enhanced DeepLabV3+ framework for automated road extraction from satellite imagery. The study introduces a Dense Depthwise Dilated Separable Spatial Pyramid Pooling module to replace the conventional Atrous Spatial Pyramid Pooling component. By integrating dense feature extraction, depthwise separable operations, and a Squeeze-and-Excitation mechanism, the proposed approach improves the representation of complex and multi-scale road structures. The contribution demonstrates the relevance of carefully designed segmentation modules for remote sensing applications, especially when road geometry is discontinuous, narrow, or embedded in visually complex urban and rural scenes.

Contribution 3 investigates automated gluten detection in bread images using convolutional neural networks. The proposed RGB approach is designed as a decision-support tool for individuals with celiac disease or gluten sensitivity, who may encounter difficulties when assessing food in social or commercial settings. The authors curate and annotate image datasets from online and custom sources and fine-tune pretrained convolutional networks for gluten-related classification. The study shows the potential of consumer-level image-based methods to complement existing gluten testing approaches while also illustrating the challenges associated with visual ambiguity, dataset construction, and transfer learning in food-related applications.

Contribution 4 introduces SEM-Net, a social–emotional music classification model designed to support emotion regulation and music literacy in individuals with special needs. The proposed model uses a convolutional neural network architecture and feature preprocessing strategies to classify musical excerpts according to positive social–emotional elements, including positive outlook, empathy, and problem-solving. The reported classification performance indicates that deep learning can contribute to educational and therapeutic contexts when domain-specific features and meaningful target categories are carefully defined. This contribution expands the scope of the Special Issue beyond visual imagery and demonstrates the broader applicability of learning-based classification in multimodal and affective computing.

Contribution 5 presents a convolutional neural network for glioblastoma, IDH-wildtype, detection from T1-weighted MRI. The study optimizes model depth, activation functions, and hyperparameters using the RSNA-MICCAI dataset and validates the model

externally on the Erasmus Glioma Database. The inclusion of external validation is particularly important because of differences in anatomical representation and acquisition characteristics between datasets. In addition, the study performs statistical analyzes to assess demographic bias in the predictions. This contribution is significant because it combines diagnostic performance with fairness-related evaluation, reinforcing the need for reliable and responsible artificial intelligence in medical imaging.

Contribution 6 develops an automated method for wall moisture detection in heritage sites using infrared imagery and convolutional neural networks. The study focuses on the Tainan Confucian Temple and addresses the time-consuming nature of manual interpretation in infrared thermography. The proposed system detects wall regions, identifies low-temperature areas, and estimates moisture distribution. By combining visible and thermal images with preprocessing and task-specific CNN models, the method supports nondestructive diagnosis in cultural heritage conservation. The study demonstrates how applied computer vision can assist preservation professionals by providing repeatable, objective, and scalable assessment tools.

Contribution 7 presents MonoLENS, a lightweight and efficient network for self-supervised monocular depth estimation. The method extends Lite-Mono by reducing computation while preserving geometric fidelity, particularly with respect to relative depth relations, object boundaries, and planar structures. The model introduces depthwise separable upsampling and multidimensional collaborative attention to sharpen structural cues while limiting floating-point operations. This contribution addresses one of the central challenges in practical depth estimation: achieving sufficient accuracy for real-world deployment without imposing excessive computational requirements on edge devices.

Contribution 8 proposes FEM-YOLO, a lightweight steel strip surface defect detection algorithm. These architectural changes aim to improve defect localization and recognition while reducing model complexity. These architectural changes aim to improve defect localization and recognition while reducing model complexity. The study is relevant to industrial inspection because surface defects may be visually subtle, spatially small, and costly to miss in production environments. The contribution demonstrates how object detection architectures can be adapted to manufacturing quality control through targeted lightweight design.

Contribution 9 introduces a real-time method for rear car signal detection in advanced driver assistance systems. The proposed approach combines a meta-learning-enhanced YOLOv8 model with geometric post-processing to detect left and right turn indicators and brake signals. The study addresses a limitation of radar and LiDAR systems, which can infer motion but do not directly interpret color-coded rear light signals. By using camera-based perception to decode visual signal patterns, the proposed method improves situational awareness and supports earlier recognition of driver intent. This contribution illustrates the role of adaptive learning and geometric reasoning in safety-oriented vehicular perception.

Contribution 10 proposes a robust self-supervised monocular depth estimation framework guided by intrinsic albedo estimation. The work addresses a key limitation of photometric reconstruction losses, namely their reliance on Lambertian reflectance assumptions. Under specular reflections or strong illumination gradients, photometric losses may lead to erroneous depth learning. The proposed multitask framework incorporates albedo estimation as an auxiliary supervised task and integrates albedo supervision into the training objective. Experiments on ScanNetV2 demonstrate improved depth accuracy without increasing inference-time parameters or floating-point operations. This study contributes to more robust self-supervised depth learning under challenging illumination conditions.

4. Conclusions and Future Directions

The papers collected in this Special Issue demonstrate that applied computer vision is increasingly characterized by domain adaptation, architectural efficiency, and concern for trustworthy deployment. Across the included contributions, several common directions emerge. First, task-specific architectural design remains essential. Generic deep learning models can provide strong baselines, but practical gains often arise from modules that reflect the structure of the target problem, such as multi-scale segmentation blocks, lightweight feature extractors, albedo-guided learning objectives, or geometric post-processing.

Second, robustness and generalization are central to applied deployment. The studies in this Special Issue address sources of variability that include thermal noise, cluttered backgrounds, lighting changes, domain shifts between imaging datasets, demographic factors in medical data, and the visual ambiguity of real-world food and heritage imagery. Future work should continue to emphasize external validation, cross-domain evaluation, uncertainty estimation, and bias analysis.

Third, efficiency is becoming a defining requirement. Several contributions explicitly seek to reduce computational cost while maintaining performance, particularly for edge devices, real-time vehicle perception, industrial inspection, and lightweight depth estimation. Future research should further integrate model compression, neural architecture search, hardware-aware optimization, and energy-efficient inference.

Finally, the Special Issue confirms that artificial intelligence in applied vision should be evaluated not only by accuracy but also by its capacity to support human decision-making. Whether used in medicine, food safety, cultural heritage, transportation, remote sensing, security, or manufacturing, these methods must be interpretable, reliable, and aligned with practical workflows. Continued collaboration among computer scientists, domain experts, practitioners, and end users will be necessary to ensure that the next generation of applied computer vision systems is both technically advanced and socially useful.

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